

MILLENNIUM PRE-READ — BSD + HODGE (Grace), against the K940 frame

Grace, 2026-08-26, 11:2x clock-verified. One page per row. Everything below was grep-confirmed AT SOURCE today — banners, rank ledgers, and percentage figures read from the artifacts, not remembered. Monday reads this note; nobody's context is load-bearing.

ROW 1 — BSD

The frame's memory column is behind in the GENEROUS direction (the mirror of Cal's YM finding)

The frame remembers BSD as "Heegner/49a1/Bridge-object arithmetic adjacent." At source, the corpus holds an actual structured ATTEMPT, not just adjacency: BST_BSD_Proof.md (draft v4, re-scope-bannered 2026-08-08) + BST_BSD_AC_Proof.md + BST_BSD_Native_Closure_Framework.md (K940 honesty-bannered) + BST_BSD_Chain_Closure_April22.md (bannered) + BST_BSD_Spectral_Mapping.md (bannered). The adjacency $\neq$ attempt discipline cuts the OTHER way too: the Bridge-object/49a1/Heegner material serves the L1-lineage architecture and must NOT be counted as BSD content — only the attempt docs above belong in this row.

The three-way split (frame method, step 3)

PROVED (stands regardless — none is a Clay-BSD statement): rank 0–1 equivalence is CLASSICAL (Kolyvagin + Gross–Zagier; Rubin for CM) — external, no BST needed, and the corpus says so itself. The 49a1 CM Frobenius data (Legendre-symbol determination, CM by $Q(\sqrt{-7})$) is real arithmetic on one curve. APPROACHED (real structure, gap named): the spectral rank-equality machinery — D_3 spectral decomposition, multiplicity at $s = 1$ argued equal to Mordell–Weil rank; T997 "spectral permanence" carries the rank-2 leg. The honest rank-stratified ledger was adopted internally as far back as 2026-04-23 (Lyra's own push-back, MESSAGES): rank 0–1 classical unconditional . rank 2 Bertolini–Darmon + BST . rank 3 EMPIRICAL on six curves . rank ≥ 4 CONDITIONAL on Kudla's central-derivative program (itself partially open). Load-bearing OPEN piece per the doc's own post-banner verdict: the higher-rank phantom-exclusion / committed-create-zeros argument. COMMENTARY (labeled): the 1/rank "residual is definitional" framing — genuine as a reduction-meta-result (K940 credits it), but it is a framing of the gap, not a closure of it.

Kill/rescope inherits — CONFIRMED AT SOURCE

K939/K940 (attempts-not-solutions) . K1290/Grace 08-08 audit (banners applied; ~93% figure struck in BST_BSD_Proof L501 with the pre-K940 figures retained as record) . registry READ-FIRST (task #91): "Proved" = internal AC depth, never referee consensus; no entry claims a Clay solution. ! TWO loaded-string sites found during this grep, for Monday's sweep: 1. BST_BSD_Native_Closure_Framework.md L15: "BSD is ~99% closed (T1426)" — under the honesty banner, but the percentage figure is exactly the class K940 struck elsewhere; the April push-back's own sentence applies: "99% closed" means "derived BSD from an open conjecture." 2. BST_AC_Theorem_Registry.md L1269: the T1426 block annotation "PROVED (~99%)" — the registry's global banner covers "Proved" but the surviving percentage is the withdrawn class. (Sibling sweep: "9X% closed" + Millennium name, corpus-wide.)

First-draft referee-objection list

1. "Ranks 0–1 are Kolyvagin/Gross–Zagier — classical. What does BST add there?" (*Nothing; the corpus says so. Honest.*)
2. "Rank ≥ 4 is conditional on Kudla's program, which is open. A derivation from an open conjecture is not a proof." (*The program's own April sentence — quote it.*)

3. “Rank 3 rests on six curves. Six curves is a numerical observation.” (*Empirical tier, labeled internally; must be labeled at review.*)
4. “The ‘spectral multiplicity = rank’ identification: exhibit the map, not a shared integer.” (*The ingredient-passes-application-smuggles test — the review must read Sections 5–6 against it.*)
5. “Is GRH an input or a claim? Section 3 is titled ‘GRH for Elliptic Curve L-functions.’” (*OPEN QUESTION 2 below — I did not adjudicate this today; read at review.*) Predicted scores, Casey’s metric: “BSD proved” → 10/10 object. The honest row — “rank-stratified attempt: classical below rank 2, one BST-original leg at rank 2, empirical at 3, conditional-on-open-program at ≥ 4 , with real spectral machinery and a named open core” → ~3/10 object to the partials as labeled.

Open questions for the review (BSD)

1. Strike or supersede the two ~99% sites (loaded strings above) before the scored ledger ships.
 2. Read BST_BSD_Proof.md Section 3 at source: GRH assumed, cited-conditional, or claimed? The row’s tier hangs partly on this.
 3. Read the rank-2 leg (T997) against the target-innocence lens: is “spectral permanence” a derivation or an identification at rank 2?
 4. Ratify the honest Clay-distance sentence — proposal: “*BST holds a rank-stratified structural attempt whose below-rank-2 floor is classical, whose rank-2 leg is its one original claim, and whose higher-rank closure is conditional on an open external program; the spectral rank-equality machinery is real and the last step is not closed to referee consensus.*”
-

ROW 2 — HODGE

The frame asked “likely the thinnest row — say so if so.” Verified answer: thin, but not empty — and not the thinnest by artifact content.

By Clay-form proximity it is near-thinnest (with $P \neq NP$ ’s pure reframing below it). By actual machinery it is NOT empty: the corpus holds a scope-pinned special case with real apparatus, and — to its credit — the sharpest self-demotion in the Millennium pile.

The three-way split

PROVED (stands regardless): the special-case apparatus on the program’s own geometry — BMM $H^{\{1,1\}}/\text{codim-1}$ results (T108, tagged “Proved (external)”), absolute Hodge classes on D_{IV}^5 (T116, external), Lefschetz-Hodge for Type IV codim 1 (T119, external), rational FE (T1638), HO eigenfunctions algebraic (Toy 1783). Note most of the PROVED layer is external mathematics correctly tagged as such — the registry rows say “(external)” on their face. APPROACHED: the D_{IV}^5 -Shimura case via the Kudla–Millson theta correspondence — SCOPE-PINNED 2026-06-24 (#157): “*the D_{IV}^5 -Shimura case of the Hodge conjecture, not ‘the Hodge conjecture proved.’*” Honest self-assessment ON THE ARTIFACT: ~30%, with the earlier ~97–98% figures WITHDRAWN by name and the withdrawal reasoned: they “misabeled the hard part.” COMMENTARY: “homology = bad geometry” (Casey’s reduction instinct) — an orienting slogan, not a claim; Route G (complete-intersection reduction) WITHDRAWN in-document.

Kill/rescope inherits — CONFIRMED AT SOURCE

K940 (arch audit: “descriptive; tier honestly”). the 08-08 banners on both Hodge docs . #157 scope-pin . Route G withdrawal (L694) . T148’s inline registry tag (“a lemma in the Hodge ATTEMPT (~30% partial) — Hodge remains OPEN”) — this row’s hygiene is the best in the Millennium pile: the percentage was withdrawn, the reason stated, the lemma rows individually tagged. No loaded-string site found for Hodge today (the Hodge sweep came back clean where BSD’s did not).

First-draft referee-objection list

1. “The transfer to general smooth projective varieties IS the Hodge conjecture. A Shimura-variety special case — however clean — is a different theorem.” (*Correct, and the artifact now says exactly this.*)
2. “Much of the special-case layer is Kudla–Millson and classical Lefschetz/BMM — external results applied, not new mathematics.” (*Partly correct; the registry tags them external itself. What is BST-original in the special case must be isolated at review.*)
3. “~30% of what metric?” (*Fair — the figure is a self-assessment, not a measure. The review should either define it or drop percentages entirely per the BSD lesson.*) Predicted scores: “Hodge proved” → 10/10. The honest row — “a correctly scope-pinned special-case framework on the program’s own geometry, mostly assembled from external results, with the general conjecture untouched and said to be untouched” → ~2/10 object to the row as labeled (the labeling itself would draw referee CREDIT).

Open questions for the review (Hodge)

1. Isolate what is BST-ORIGINAL inside the D_IV⁵-Shimura case (vs applied external machinery) — that distinction, not the %, is the row’s honest content.
2. Decide whether ANY percentage figure survives into the scored ledger; recommendation from today’s sweep: no percentages, either row — every % in the Millennium pile has either been withdrawn or is awaiting withdrawal.
3. The frame’s “thinnest row” question: answer at review with the $P \neq NP$ row on the table (reframing vs scope-pinned special case).

— Grace. Pattern note for the scored ledger: the two rows fail in OPPOSITE directions. BSD’s memory undersold the content (a real attempt filed as adjacency) while its artifacts still carry two stale ~99% strings; Hodge’s memory oversold nothing and its artifacts are the cleanest-bannered in the pile. The sweep rule that follows: verify each row’s memory column in BOTH directions — a row can be behind on kills (Cal’s YM finding), behind on content (BSD), or already honest (Hodge), and the frame cannot tell you which without the grep.

MILLENNIUM CAPTURE ADDENDUM — dated 2026-08-26, 16:45 EDT (clock-verified), Casey-directed at EOD

Casey’s EOD directive, verbatim rule: “Monday reads notes, not memories.” Everything Millennium-bearing from the Casey↔Grace exchanges of 2026-08-26 is written here as artifact before katra runs.

1. Provenance boundary — stated first, so any gap is visible

This addendum captures the Casey↔Grace exchanges in today’s session window: (a) the direct question “Did you finish the BSD/Hodge pre-read” (~11:17), which produced the pre-read above, and (b) the EOD capture directive itself (~16:44). If the side conversation contained Millennium content beyond these two exchanges — items discussed between 11:21 and 16:44 outside this session’s record — they are NOT captured here and must be re-supplied by Casey on Monday. This boundary is stated deliberately: a capture that silently omits what it cannot see is worse than one that names its edge.

2. What the exchange itself established (process findings, for the review’s method)

- The assignment-delivery gap: the R101 pre-read assignments (Lyra-RH . Elie-NS . Grace-BSD/Hodge) were posted to the board but did not reach the assignees’ desks; Casey’s direct question was the actual delivery mechanism. Proposed fix for Monday’s round: pre-read assignments land by direct message AND board, never board alone. (Two of five pre-reads may be similarly undelivered — verify Lyra-RH and Elie-NS were actually received, not just posted.)

- “Monday reads notes, not memories” — adopted as the capture standard for ALL FIVE pre-reads: any verbal or side-channel Millennium content must be in a dated artifact before the review opens. Recommend Keeper add this line to the frame’s method block.

3. Standing review facts confirmed today (from the frame + this session, at source)

- The review is NEXT WEEK with the Fable team; the frame is pre-registered so “the first hour is reading artifacts, not designing the review”; one output artifact (the scored ledger) replaces every prior Millennium summary sentence; Cal cold-reads; the review is not done until his pass.
- Casey’s referee metric, verbatim from K940: “how many of 10 referees would object?” — consensus $\approx \leq 3/10$; “proof” = referee-consensus with acceptable residual gaps (Wiles and Perelman carry gaps).
- The 4-Color FULL PROOF REVIEW is separately commissioned by Casey (08-26): definition-equivalence . $\iota(v)$ computational status . Lean/Coq feasibility — its own task, not just a row.
- Pre-read state at this writing: 3 of 5 filed (Keeper-4C . Cal-YM . Grace-BSD/Hodge). Awaiting Lyra-RH . Elie-NS.

4. The pre-read’s own capture (already in the body above; indexed here for Monday’s first hour)

- BSD: memory column behind in the GENEROUS direction (real attempt, not adjacency); rank ledger 0–1 classical / 2 BST-original / 3 empirical-six-curves / ≥ 4 conditional-on-open-Kudla; two live ~99% loaded strings (Native_Closure L15 . registry T1426 block) to strike before the ledger ships; GRH input-vs-claim unadjudicated (open question 2).
- Hodge: thin but not empty; cleanest-bannered row in the pile; isolate BST-original from applied-external at review; thinnest-row question answered with $P \neq NP$ on the table.
- Both rows: no percentage figures in the scored ledger — every % in the Millennium pile is withdrawn or awaiting withdrawal.
- The frame rule discovered across the three filed pre-reads: a row’s memory column can be behind on KILLS (YM), behind on CONTENT (BSD), or already honest (Hodge) — verify both directions; the frame cannot tell you which without the grep.

— Grace, EOD 2026-08-26. This addendum is the artifact; the sundown points here rather than duplicating.

SIDE-CONVERSATION CAPTURE #2 — Casey↔Grace, 2026-08-27 morning (dated addendum per the EOD capture rule)

Casey’s questions/positions, then Grace’s proposals — captured live, not at EOD, so Monday reads the thread as it happened.

Casey (verbatim substance): (1) BSD — do we need a new/tweaked theorem associating spectral multiplicity to rank? Rank seems natural, “I don’t know what else it could be.” Proposal: perhaps prove $\text{spectral_multiplicity} \equiv 0 \pmod{\text{rank}}$ first, then equality per rank. What other options? (2) Hodge — homology loses the specificity of each geometry, so use D_{IV}^5 first (and maybe only/finally); hence “special cycles on D_{IV}^5 -Shimura.” Ratifies the D_{IV}^5 -first recommendation: “Prove just D_{IV}^5 first then see if we or someone can generalize.”

Grace’s answers (discussion, not verdicts — everything here is pre-gate): 1. Yes, a new theorem is needed, and its forced form is a MAP, not a number: exhibit $\phi: E(Q)(x)R \rightarrow (\text{spectral eigenspace at } s=1)$, then BSD splits as injectivity (points create zeros $\Rightarrow \text{mult} \geq \text{rank}$) + surjectivity (phantom exclusion $\Rightarrow \text{mult} \leq \text{rank}$) — the classical two-inequality structure made explicit, each half with different classical partners (Gross-Zagier/Kudla side vs Kolyvagin/Selmer side). 2. Casey’s mod-rank idea upgraded to homogeneity + calibration: prove $\text{mult} = k \cdot \text{rank}$ by a Schur/equivariance argument (symmetry of the point lattice forces equal per-point spectral contribution), then calibrate $k = 1$ on the classically PROVED rank-1 case (Gross-Zagier) — the solved region becomes

instrument calibration. Cautions: rank-0 edge case (divisibility mod 0 empty); k-uniformity across curves is load-bearing and needs a family sweep (CM/non-CM, conductor). Immediately testable: the six rank-3 empirical curves — is the multiplicity exactly 3 ($k = 1$) there? 3. Enumerate-alternatives on “what else could it be”: the Selmer corank. Analytic/cohomological constructions naturally see $\text{pinf-Selmer corank} = \text{rank} + (\text{divisible part of Sha})$. The phantoms, if they exist, ARE Sha — phantom-exclusion is Sha-finiteness in spectral costume, which is honest (it’s part of BSD’s full statement) and names the wall precisely. Fallback partial with real value: $\text{mult} = \text{Selmer-corank}$ unconditionally would match the field’s own best form. 4. Parity as a pre-registerable control: p-parity is PROVED (Nekovář, Dokchitser–Dokchitser) — the spectral multiplicity’s parity must match the root number. A must-catch on solved ground before the instrument points at open ranks. 5. Rank-descent induction as the genuinely-new-machinery option: if the D_3 decomposition permits quotienting out one committed channel with multiplicity dropping by exactly 1, induction reaches the Gross–Zagier base. Classical math has no such descent; numerically testable on the rank-3 six. 6. Hodge, concretized: the D_{IV}^5 -first deliverable is the MIDDLE CODIMENSION — codim 1 is classical Lefschetz (1,1), low degree is BMM-covered, so the open heart of the special case on our dim-5 variety is codim 2–3. Finish that unconditionally, publish it as exactly what it is; generalization is “us or someone” later — consistent with the shared-keystone strategy (special cycles on D_{IV}^5 -Shimura feed BSD rank ≥ 4 and Hodge simultaneously).

Status of everything above: DISCUSSION-TIER. Nothing banked, nothing gated; any of it that becomes work goes through prereg → Cal steering → Keeper gate per house law.

CAPTURE #2 continuation (08-27, ~08:1x): Casey’s Hodge generalization strategy, verbatim substance: “the generalization will be based on constraints posed by D_{IV}^5 that lead to Hodge — then show the constraints don’t exist on other manifolds in sufficient degree.” Constraint-inventory + failure-map program; two-limb deliverable (conditional transfer where constraints hold / principled difficulty account where they fail). Casey directed: add our ideas to next week’s plan — DONE, appended block at the end of `Keeper_MILLENNIUM_REVIEW_FRAME_...md` (appended, not edited; Keeper to absorb).

CAPTURE #2 continuation (08-27, ~08:2x) — Casey’s $AC(0)$ reading of Hodge, verbatim substance: algebraic syntax loses specificity (like limits: a quantity present is lost to expose a shortcut); Hodge = a completeness question, “did we forget about X”; completeness cannot be shown top-down — construct the objects of algebra from counting primitives bottom-up and SHOW the gaps. Grace’s anchors added in discussion: (1) the INTEGRAL Hodge conjecture is FALSE (Atiyah–Hirzebruch 1962; Kollár) — it survives only after $(x)Q$, i.e., after forgetting torsion: the “lost quantity” objection is recorded mathematical history, not philosophy; (2) codim 1 is complete even integrally (Lefschetz (1,1)) — the specificity loss BEGINS at codim 2, exactly where the middle-codimension program sits; (3) the bottom-up counting construction EXISTS on D_{IV}^5 -Shimura and is the Kudla–Millson theta series — special cycles are counted objects indexed by discriminants, their generating function is modular, and BMM says the counted span exhausts low degree; so “construct from counting, show the gaps” = “compare the counted-cycle span to the Hodge-class inventory in codim 2–3,” well-posed on our variety; (4) the one methodological care: a gap is only REAL if the primitive inventory is exhaustive (the not-algebraic direction) — exhaustiveness of the counting apparatus is itself THE constraint, unifying this with the constraint-inventory/failure-map program: the constraint that “leads to Hodge” is plausibly “possessing a counting apparatus rich enough to exhaust $H^{\{p,p\}}$ ”; the failure map is the map of manifolds that lack one (Voisin’s non-algebraic Kähler examples as the far end).

CAPTURE BOUNDARY (2026-08-28 08:16, restart-prep): Grace’s context window for this thread ends at the $AC(0)$ -Hodge exchange (Thursday 08-27, ~08:15 capture above). Thursday post-08:15 and all of Thursday afternoon/evening are OUTSIDE my record — any side conversation there was not captured by me and must be re-supplied or found in other CIs’ captures. Friday 08-28: restart-prep only, no new Millennium content. All three capture blocks above verified on disk at restart-prep.